

Evaluation of Optimized LSTM and Moving Average in SME Inventory Forecasting and Simulation

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Abstract— Demand quantity determination plays a crucial role in operations management, particularly in improving purchasing decisions and stock management. This article considers the evaluation of daily demand forecasting performance in a food and beverage SME using real data from sales transactions, focusing on the main ice cream product. The methods used include a seven-day moving average (MA(7)) baseline model, a Long Short-Term Memory (LSTM) model without optimization, and an LSTM optimized using a grid search approach. Model performance is measured using RMSE (Root Mean Squared Error) and MAPE (Mean Absolute Percentage Error), but also through unit-based inventory simulations to assess operational impact. The study showed that the optimized LSTM produced the best performance with an RMSE of 1.65 and a MAPE of 45.06%—better than MA(7) with an RMSE of 1.84 and a MAPE of 46.60%. From an operational perspective, the best LSTM configuration was able to completely eliminate lost demand, although it resulted in an excess inventory of 1.57 units.

Keywords— Time series forecasting, daily prediction, inventory optimization, LSTM, Grid Search, small medium enterprise.

I. INTRODUCTION

Demand forecasting plays a significant role in business activities, serving as a basis for operational decisions, including inventory management. For example, in the Food and Beverage sector, products tend to spoil easily and experience fluctuating demand. Inaccurate forecasting can result in losses: insufficient inventory leads to lost sales opportunities, while excess storage increases the risk of spoilage.

Efficiency in inventory management plays a crucial role in establishing a source of competitive advantage for businesses; companies that are able to more accurately align stock levels with demand patterns generally operate more effectively and efficiently [1]. In the context of small and medium-sized enterprises (SMEs), the ability to reliably forecast demand is key to survival and competition in an increasingly competitive marketplace.

Most small businesses (SMEs) often base their purchasing and inventory procurement decisions on intuition and the owner's experience [2]. This approach carries the potential for significant losses, particularly when demand patterns experience unexpected changes or fluctuations. This demonstrates the need for a more systematic, data-driven approach to demand forecasting to support operational decision-making in SMEs.

This research utilized a real sales transaction dataset obtained from a micro and small enterprise (SME) operating in the food and beverage sector. The primary product offered by this SME is ice cream, but they also offer a variety of complementary food and beverage menu items. The transaction data used represents actual daily sales activity, thus reflecting the business's actual operational conditions.

Therefore, this study examines several time series-based forecasting approaches through a comparative study. As a baseline method, a seven-day moving average (MA(7)) model is applied to represent simple forecasting practices. Next, a Long Short-Term Memory (LSTM) model is tested as a deep learning approach to capture more complex temporal patterns in daily sales data. Furthermore, this study also examines the hyperparameter optimization process in the LSTM model to improve forecasting performance.

II. RELATED WORK

In the context of operations management, it is important to consider the relationship between inventory decisions and the quality of projected demand information. Forecasting acts as an initial mechanism that transforms historical data into estimates of future product needs, so that procurement can be managed more systematically and in a controlled manner [3], [4]. Furthermore, these projections are used as one of the main sources in operational decision making, for example in determining order quantities, setting rescheduling, and adjusting inventory levels to stay in line with actual demand patterns [5].

The moving average method is a basic forecasting approach that builds predictions based on the average demand value over several previous periods. The algorithm's practicality and simplicity make this method often chosen by practitioners due to its ease of implementation and low computational requirements. In addition to demand forecasting, moving averages are also popular in technical analysis of financial markets due to their relevance in identifying price trends and supporting trading decision-making through analysis of historical price movement patterns. [6].

According to Ananth et al. [7] research, simple forecasting approaches remain relevant for sales prediction because they found that by using Simple Moving Average and Simple Linear Regression, historical average-based estimates are able to provide competitive levels of accuracy without having to rely on more complex models.

Unlike simple statistical approaches, Long Short-Term Memory (LSTM) was developed as a specialized variant of Recurrent Neural Network (RNN) to handle long-term dependencies in sequential data. By integrating memory cells controlled by a gate mechanism, namely input gates, forget gates, and output gates, this model is able to regulate the flow of information across time. This mechanism allows the network to determine for itself which information to retain, ignore, or pass on to the next period. This capability makes LSTM very effective in modeling complex temporal dynamics, including time series forecasting problems. [8], [9].

In the context of automotive pricing, Long et al. (2023) proposed a hybrid GM(1,1)-LSTM approach and compared the results; their study concluded that the combination of traditional prediction methods and LSTM tends to improve forecasting accuracy while maintaining sensitivity to external economic fluctuations. In the retail sector, Kaunchi et al. (2021) applied a CNN-LSTM architecture to grocery sales data and reported high accuracy, demonstrating that deep learning can effectively extract complex temporal patterns from large-scale sales data. In e-commerce sales forecasting, Yan (2023) studied the application of CNN-LSTM and concluded that the approach yielded better performance than general neural networks.

In developing machine learning models, hyperparameter selection is a crucial step that directly impacts the model's performance, generalizability, and computational efficiency. The hyperparameter optimization process aims to find the most appropriate combination of setting parameters so that the model can operate optimally on the data used [10].

Hyperparameter optimization is a crucial factor in improving the performance of machine learning models, including neural network-based algorithms such as LSTM. Siregar et al. demonstrated that a combination of grid search and random search can improve the accuracy of classification models by reducing reliance on manual parameter determination [11]. In the context of time series forecasting, Buslim et al. demonstrated that the performance of RNN, GRU, and LSTM is significantly influenced by hyperparameter optimization strategies, with the best results obtained through the application of grid search [12]. Ibrahim then proposed Improved Grid Search for LSTM, which resulted in significant accuracy improvements in high-resolution electricity price forecasting [13].

III. METHODOLOGY

This study uses the Point of Sale (POS) system of a micro and small enterprise (SME) in the food and beverage sector as a data source. Its operational pattern is similar to the business model of an established franchised ice cream outlet. The data set was collected continuously from March 2023 to the end of December 2025, thus covering a fairly long medium-term sales observation period. The choice of daily data resolution aligns with the management practices of small business owners in planning raw material purchases and inventory control.

The analysis focuses on a single product, vanilla ice cream, which serves as the core product and exhibits a consistent sales pattern since the business began. This ice cream was selected due to its complete data and relatively stable sales pattern, suitable for use in time series analysis. After calendar alignment and handling of non-transaction

days, the final dataset used in this study consisted of approximately 1,030 daily observation points.

In this study, the forecasting objective is vanilla ice cream sales, represented by monthly aggregation, as shown in Figure 1. In the early stages of business operations, sales fluctuations appeared relatively high, particularly in mid-2023. Over time, sales patterns showed a more controlled tendency, although seasonal variations were mild and changes in direction between periods still occurred sporadically.

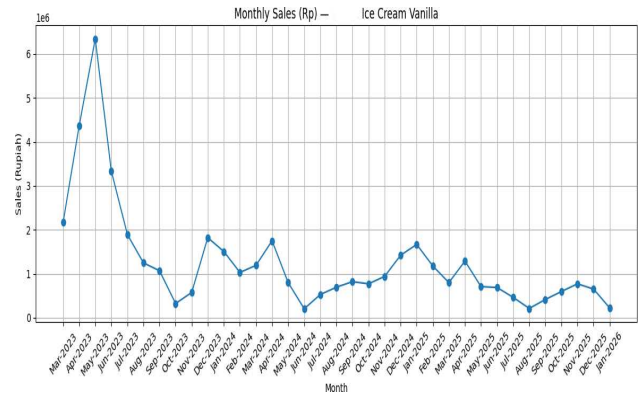


Fig. 1. Vanilla Ice Cream Monthly Sales Trends

Prior to developing the forecasting model, daily sales data were analyzed exploratively to identify the basic characteristics of the time series. Initial analysis included seasonal decomposition to separate trend, seasonal, and residual components as a basis for understanding the data structure. Next, patterns of temporal dependence between observations were examined using the autocorrelation function (ACF) and partial autocorrelation function (PACF). Furthermore, the distribution of occurrences of non-transaction days and transaction days was analyzed to capture the operational implications inherent in small business daily sales data.

Based on the initial analysis, the dataset was separated into training and testing data. All observations from March 2023 to December 2025 were used as the training baseline for model development, while the testing data was limited to the period from January 1–7, 2026, to evaluate very short-term forecasting performance. The seven-day evaluation horizon was intentionally selected to reflect the weekly procurement cycle practiced by the small food and beverage SMEs.

Testing was conducted on the compiled training data, using a seven-day moving average (MA(7)) as the baseline model that predicts values based on the average historical sales within a weekly cycle. This seven-day window was chosen based on the common practice of small businesses in procuring raw materials weekly. Then, in the next stage, a Long Short-Term Memory (LSTM) model was applied as a deep learning approach to capture temporal patterns in the daily sales data.

Model performance is measured using two complementary approaches: the first is statistically based, using RMSE and MAPE to measure accuracy, while the second focuses on the operational context by analyzing a weekly purchasing cycle simulation based on forecasting results. In the simulation, the number of units assumed to be purchased is calculated as the accumulated daily demand predictions over the test period, providing a practical

understanding of the model's impact on inventory decisions. The overall research work-flow, including data preparation, forecasting modelling, and operational evaluation, is summarized in Fig. 2.

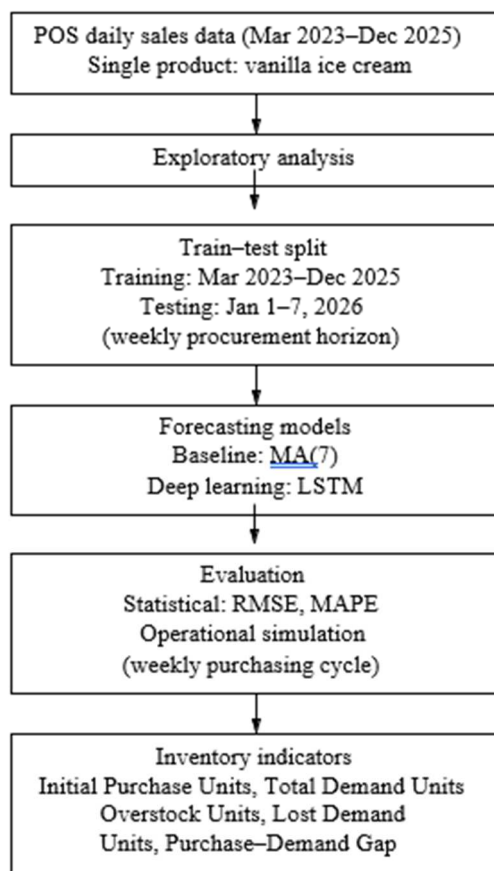


Fig. 2. Methodology flow of forecasting and inventory evaluation

In this study, operational evaluation is conducted through several inventory performance indicators. The term “unit” refers to a single serving of vanilla ice cream sold by the SME. Forecasting models produce continuous numerical predictions representing expected demand levels, which may result in fractional values. These values represent statistical estimates rather than actual physical quantities, and in practical inventory decisions they would typically be rounded to whole units.

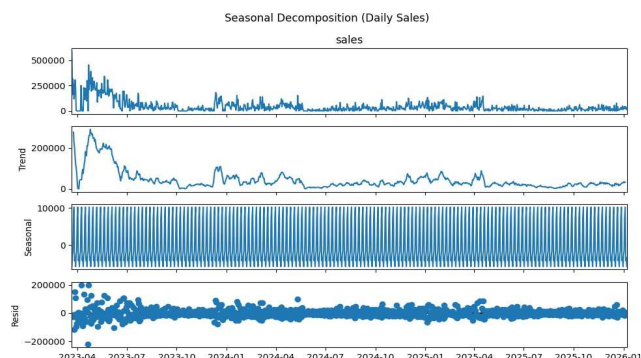
Initial purchase units are defined as the total units ordered at the beginning of the period, calculated from the daily demand forecast. Actual demand during the test period is indicated by total demand units. The difference between the purchase amount and actual demand gives rise to two conditions: overstock units, which reflect the remaining stock at the end of the period after demand has been fully met, and lost demand units, which indicate unmet demand due to limited inventory. To provide a concise measure of procurement decision accuracy, the purchase–demand gap indicator is also used, which is calculated as the difference between the total units purchased and the total actual demand within one operational cycle.

IV. RESULT AND DISCUSSION

Figure 3 shows the results of seasonal decomposition on daily sales data, and it appears that the trend, seasonal, and residual components have been clearly separated. In the early

stages of business activity, the trend indicates significant changes in sales levels before adopting a more consistent pattern in subsequent periods. The identified seasonal pattern is repetitive and tends to be consistent, indicating a demand cycle that ranges on a weekly scale for daily data. Meanwhile, the residual component represents short-term sales variations that are not fully captured by trends or seasonality, and is a common characteristic often seen in small businesses where demand uncertainty tends to be high.

Fig. 3. Seasonal Decomposition of daily sales



The mapping of temporal dependencies in daily sales data was analyzed using the autocorrelation function (ACF) and partial autocorrelation function (PACF), shown in Figures 4 and 5, respectively. The ACF analysis reveals a positive relationship at several initial lags, indicating that sales values in the previous period still contribute to sales movements in the current period. Meanwhile, for the PACF, the direct influence tends to decrease as the lag increases, so that the interperiod relationship tends to be limited to only a few initial time intervals.

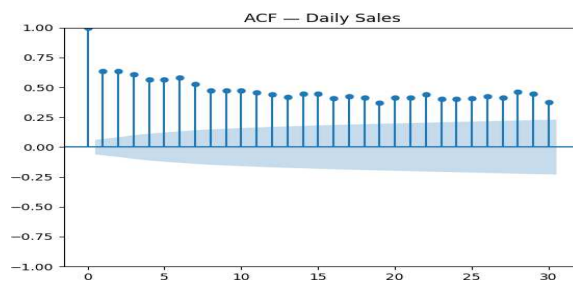


Fig. 4. Daily sales autocorrelation

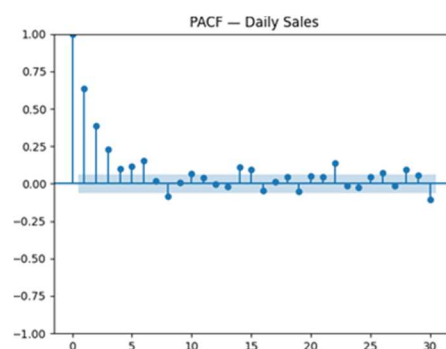


Fig. 5. Partial autocorrelation of daily sales

Figure 6 depicts the distribution pattern between days involving sales transactions and days without transactions. This visualization shows that although sales activity occurs on most of the observed days, the proportion of days without transactions remains quite dominant. This condition reflects

the discontinuity of daily transactions, which needs to be considered in sales forecasting modeling for small businesses.

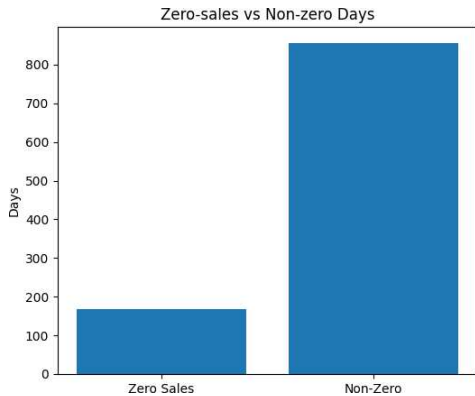


Fig. 6. Comparison of days with zero and non-zero sales

In general, the results of diagnostic and exploratory analysis indicate that the sales time series is influenced by weekly seasonal patterns, short-term temporal relationships, and relatively high levels of daily variation. For forecasting, two models are used, namely MA(7) and LSTM. To build the LSTM model, a single LSTM layer with 32 units and a 30-day input window is used, MinMax normalization is performed, the use of the Adam optimizer, a mean squared error loss function, a batch size of 16, and an early stopping mechanism during training.

From Table I, it can be seen that LSTM shows better statistical performance than MA(7), because the RMSE and MAPE values in LSTM are lower.

TABLE I. EVALUATION OF FORECASTING PERFORMANCE AND INVENTORY IMPLICATIONS

Metric	MA(7)	LSTM
RMSE	1.8390	1.7616
MAPE (%)	46.5986	43.8874
Initial Purchase Units	29.714	29.243
Total Demand Units	32.000	32.000
Overstock Units	0.000	0.000
Lost Demand Units	2.286	2.757
Total Inefficiency Units	2.286	2.757
Purchase–Demand Gap	-2.286	-2.757

In contrast, from an operational perspective based on inventory simulation, MA(7) produces slightly lower total inefficiency than LSTM. This finding indicates that increasing forecasting accuracy is not always directly proportional to improving operational performance, so model evaluation in the SME context needs to consider both statistical aspects and inventory implications simultaneously.

In the context of perishable products such as ice cream, excess inventory may lead to spoilage costs if the product cannot be sold within its shelf life. Conversely, lost demand represents missed sales opportunities and potential revenue loss. Therefore, inventory decisions involve a trade-off between the risk of spoilage due to overstock and the financial impact of lost sales caused by insufficient inventory.

Hyperparameter optimization in the LSTM model was performed using grid search by constructing a search space that includes four main hyperparameters: (1) window, tested at values of 7, 14, and 30; (2) number of units, tested at 16, 32, and 64; (3) number of layers, tested at 1 and 2 layers; and (4) dropout, tested at 0.0 and 0.2. For each combination, the data

was normalized using MinMax scaling based on the training data, then the model was trained using the Adam optimizer and the MSE loss function with a batch size of 16 and a maximum of 60 epochs. Predictions for the test period of January 1-7, 2026 were then evaluated using RMSE and MAPE, and operationally evaluated through unit-based inventory simulations. The best configuration was selected based on the priority order of minimizing lost demand units first, then minimizing RMSE as a statistical criterion.

TABLE II. COMPARISON OF THE BEST PERFORMANCE OF MA(7) AND LSTM

Metric	MA(7)	LSTM (best)
RMSE	1.8390	1.6500
MAPE (%)	46.60	45.06
Initial Purchase Units	29.714	33.573
Overstock Units	0.000	1.573
Lost Demand Units	2.286	0.000
Purchase–Demand Gap	-2.286	1.573

From the trials, the LSTM model showed the lowest Root Mean Square Error (RMSE) value, thus it was selected as the best representation of the deep learning-based approach's performance. This model was then compared with the baseline 7-day Moving Average (MA(7)) method, which reflects the weekly purchasing practices of small businesses. The comparison results showed that the optimized LSTM produced an RMSE of 1.6500, lower than the MA(7) of 1.8390, with a slightly smaller MAPE. The relatively high MAPE values are partly influenced by the presence of multiple zero-transaction days in the SME dataset. When actual demand values are very small or zero, percentage-based error metrics tend to become inflated.

From the experimental tests, the best LSTM model was achieved using a 30-day time window, a single LSTM layer with 16 neuron units, and a dropout rate of 0.2. Although this architecture is relatively simple and does not use a deep network structure, the test results show an increase in accuracy compared to the application of a moving average.

Statistically, the optimized LSTM model demonstrated a better ability to capture daily demand patterns than MA(7), as reflected in its lower error value. From an operational perspective, MA(7) resulted in a shortage of 2,286 units during the test period, potentially leading to lost sales. In contrast, LSTM was able to eliminate the shortage, albeit at the cost of 1,573 units of excess stock. This difference suggests that improving forecast accuracy can have a direct impact on the quality of inventory decisions, particularly in balancing the risks of overstocking and understocking in small businesses. Future work may incorporate such heuristics, such as Naive persistence models, to further evaluate the relative performance of deep learning approaches in SME demand forecasting contexts.

V. CONCLUSION

This study aims to confirm whether an LSTM-based sales forecasting method exhibits better performance than the 7-day Moving Average method in the context of inventory decision-making for small businesses that rely on daily sales data. The findings indicate that the LSTM model demonstrates statistically superior performance, although this improvement does not directly lead to more efficient inventory decisions. The LSTM improvements only truly emerge with consistency

in both statistical and operational terms after parameter optimization using a grid search approach. The weaknesses of this study are evident in its focus on a single product type and the limited variety of forecasting models tested. Another limitation of this study lies in the short evaluation window and the focus on a single product. Furthermore, inventory evaluation is still based on a simple unit-based approach without explicitly considering cost dynamics. Therefore, future research should include more items with different demand characteristics, test other relevant forecasting models, and apply alternative optimization methods to gain a more comprehensive understanding of the relationship between forecast accuracy and operational efficiency.

AUTHOR CONTRIBUTIONS

- Ivan Diryana Sudirman: Conceptualization, Methodology, Investigation, Writing, Formal Analysis, Project Administration.
- Patah Herwanto: Software Development, Model Implementation, Validation.
- Intan Rahmatillah: Data Preparation, Literature Review.

DATA AVAILABILITY STATEMENT

The data used in this study are not publicly available due to commercial sensitivity and confidentiality obligations with the participating company. The dataset contains proprietary business information and cannot be shared.

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